



ELECTRODIAGNOSIS: REPETITIVE NERVE STIMULATION (RNS)

- **Decremental response**
- **Incremental response**
- In MG → Universal decremental response

TREATMENT

00:45:40

- DOC: **3,4-diamino pyridine** → Blocks K^+ channels
 - Increases the amount of ACh at NMJ
- Plasmapheresis: Clears the Anti P/Q antibodies
- Pyridostigmine: Inhibits the breakdown of ACh

Important Information

- Honey to neonate → Botulism
- Toxin: Inhibits ACh release at NMJ → floppy baby
- Electro Dx: RNS - High frequency stimulation: Incremental response



Q. A female patient presents with drooping of eyelids at the end of the day, though she is not sleepy. She feels weak all the time. On examination, DTR and superficial reflexes are normal. ACh R antibody is detected. Which of the following is the cause of this disorder?

- a. Collagen vascular disorder
- b. Thymic hyperplasia
- c. Sarcoidosis
- d. Eosinophilic granuloma

Ans: (b)



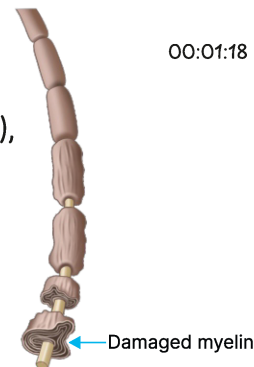
12. GUILLIAN BARRE SYNDROME



GUILLIAN BARRE SYNDROME

PATHOGENESIS OF GBS

- Autoimmune polyradiculopathy (damage at nerve roots)
- Antibody- **IgG Anti GM1** (causes Acute Inflammatory Demyelinating Polyradiculopathy), **IgG Anti GQ1b** (Miller Fisher syndrome)
 - Causes **patchy segmental demyelination** of nerve roots
 - Complement mediated activation
 - Demyelination - flaccid paraplegia → **ascending symmetrical flaccid paralysis**, conduction block
 - Recovery in 85%
 - Axonal degeneration - axons disconnected from NMJ (Neuromuscular junction)
 - Slow recovery



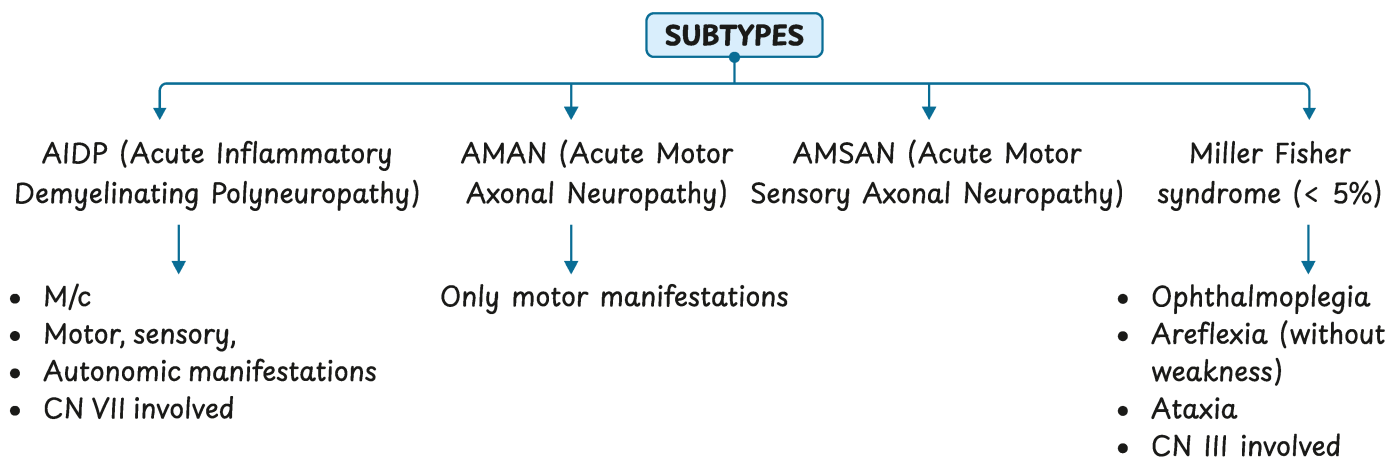
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CAUSES

- **Campylobacter jejuni** - food poisoning
- E. coli/Coliform in water
- Mycoplasma pneumoniae → walking pneumonia
- CMV/Herpes/ EBV/ HIV/**Zika virus**/ Hepatitis E
 - Vertical transmission of Zika virus - Microcephaly
 - Hepatitis E - leading cause of Acute viral hepatitis
- Influenza vaccines, older generation of neural rabies vaccine
- SARS-COV-2 : Vaccines may cause GBS
- Lymphoma, SLE - Risk of GBS incidence ↑

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SUBTYPES OF GBS



00:10:12



Important Information

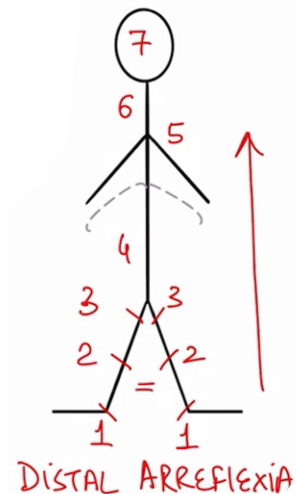
- Wernicke's encephalopathy
- Global confusion
- Ophthalmoplegia (CN VI palsy)
- Ataxia
- T/t - I/V Thiamine + I/V Dextrose

	FEATURES	NEURAL TARGET	ELECTRODIAGNOSTIC STUDY FINDINGS
AIDP	• M/c • Seen in adults • Rapid recovery • IgG Anti GM1 Ab	Schwann cells	Demyelination
AMAN	• Seen in children, young adults	Nodes of Ranvier	Axonal degeneration
AMSAN	• Seen in adults • Uncommon	Nodes of Ranvier	Axonal degeneration
MFS	• Triad - OAA • Anti GQ1b antibody	Schwann cells	Either of two

- Electrodiagnostic studies - Nerve Conduction Velocity → ↑ latency (slow)

CLINICAL MANIFESTATIONS

- Few days to week after recovery from C. jejuni food poisoning :
 - Weakness in legs → rubbery legs and instability
 - Progression of weakness : 12 hours - 28 days
 - Ascending symmetrical flaccid paralysis
 - Diaphragmatic paralysis/weakness - ventilatory support may be required, ↓ FVC, ↓ single breath count
 - Earliest manifestation - distal areflexia
 - Truncal paralysis, Quadriplegia
 - Neck floppiness
 - M/c CN involvement in GBS - CN VII (Facial diplegia)
 - Bulbar palsy : absence of gag reflex and cough reflex
 - Difficulty in handling secretions
 - Can't talk, can't eat, can't drink
 - Intubation required
 - Proprioception loss/ dysesthesia, pain in neck, arm and shoulders like in over exercised muscles
 - Sensory involvement
 - Bladder/ bowel - Spared
 - Autonomic involvement

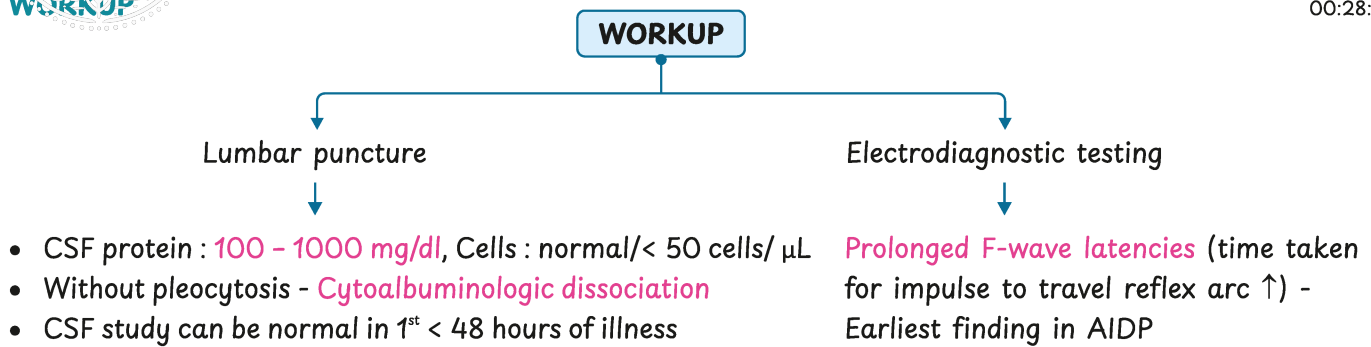


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- Wide fluctuations in BP
- Arrhythmia, Bradycardia, postural hypotension

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BRIGHTON CRITERIA (LEVEL 1 OF DIAGNOSTIC CERTAINTY) Mnemonic -FAME

00:31:56

- Flaccid B/L weakness of limbs
- Cytoalbuminologic dissociation, Areflexia ascending symmetrical
- Monophasic illness - 12 hrs - 28 days
- Electrodiagnostic findings of demyelination
- MFS (Miller Fisher Syndrome) - Areflexia without weakness, ophthalmoplegia (CN III palsy)

T/t

00:35:13

- IVIG - TOC
- PLEX/ plasmapheresis } Any of two can be used
 - Equally effective
 - Functional improvement occurs > 1 week of T/t or delayed response
- Glucocorticoids not useful
- 85% - Full recovery

CAUSE OF DEATH - 2° PULMONARY INVOLVEMENT



CIDP (CHRONIC INFLAMMATORY DEMYELINATING POLYNEUROPATHY)

- Leg weakness → arm weakness (acute onset)
- Deterioration > 9 weeks of onset or relapse > 3 times
- T/t - IVIG + Steroids

ACUTE PHARYNGEAL CERVICOBRACHIAL NEUROPATHY

- Anti GT1a Ab
- Rare form of GBS
- Arm weakness along with pharyngeal muscle weakness

MULTIFOCAL MOTOR NEUROPATHY WITH CONDUCTION BLOCK

- IgM anti GM1 Ab



Q. Which of the following best describes the characteristic triad of symptoms seen in Miller Fisher syndrome?

- a. Areflexia with muscle weakness, ataxia, ophthalmoplegia
- b. Areflexia without muscle weakness, ataxia, ophthalmoplegia
- c. Ataxia, incontinence and cortical dementia
- d. Ataxia, incontinence and subcortical dementia

Ans: (b)



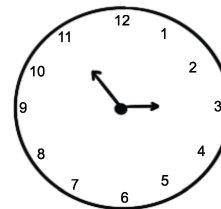
13.9 ALZHEIMER'S DISEASE

CLOCK FACE TEST

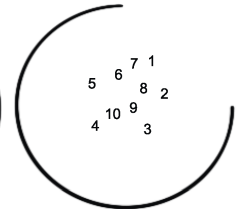
- Instruct the patient to mark the time 11:10
- Used in the diagnosis/ **evaluation of dementia**



Normal



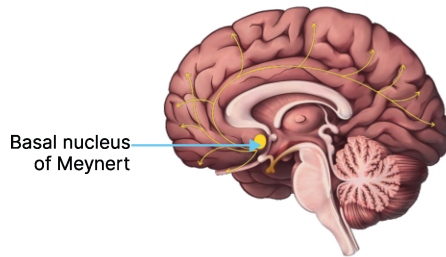
MCI, early AD



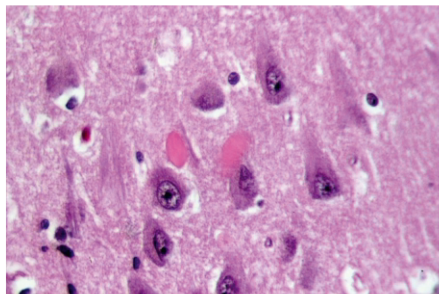
Late AD

ALZHEIMER'S DISEASE

- Basal nucleus of Meynert present in the basal forebrain is affected
- **Acetylcholine deficiency**



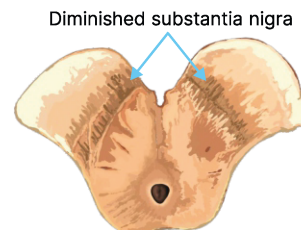
Area involved in Alzheimer's Disease



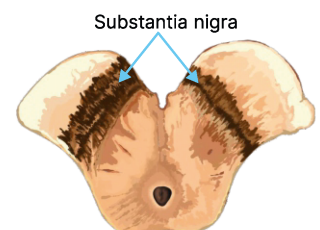
Alzheimer's disease- Hirano bodies

PARKINSON'S DISEASE

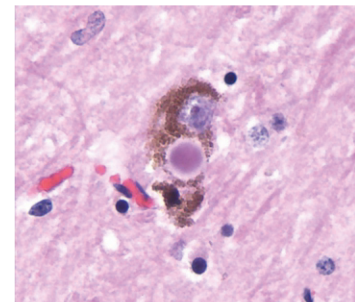
- Substantia nigra is affected
- **Dopamine deficiency**



Parkinson's disease



Area involved in Parkinson's disease

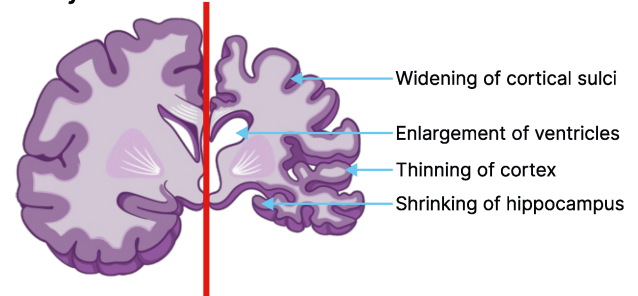


Parkinson's disease - Lewy bodies

ALZHEIMER'S DISEASE

Healthy brain

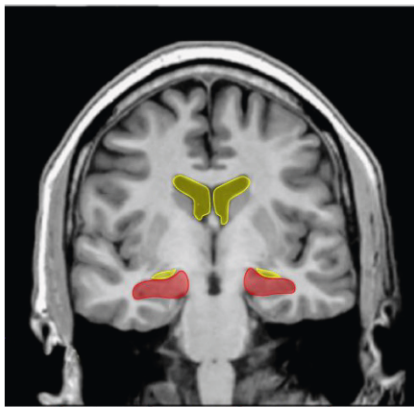
Alzheimer's brain



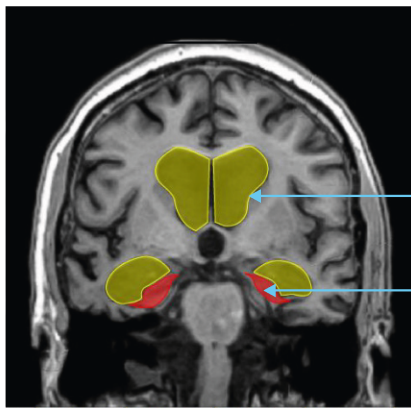
- Cortical atrophy → Thinning of the gray matter of brain → enlargement of ventricle



Normal



Alzheimer's Disease



Ventricle
Enlargement

Volume Loss of
Hippocampus

Important Information

Subcortical dementia

- Basal ganglia involved
- No impairment in recognition of family members

- M/c cause of cortical dementia
 - No basal ganglia involvement
 - Impairment of recognition of family members
- Other causes of cortical dementia
 - Alzheimer's disease
 - Vascular dementia
 - Fronto temporal dementia : **Pick's disease**
 - Variant Creutzfeldt-Jakob Disease : **Dementia + startle myoclonus**

LEADING CAUSE OF DEATH IN AD:

- **Pneumonia** (M/c) : for Both PD and AD
- Malnutrition
- Pulmonary embolism
- Heart disease
- Secondary infection

PRECLINICAL AD

- CSF, PET biomarkers evidence is present in the absence of clinical features. This can predate development of AD by 2 decades
- CSF biomarker
 - **$A\beta_{42}$ levels $\downarrow$**
 - **$A\beta_{42} / \beta_{40}$ levels $\downarrow$**
- Evidence with imaging
 - FDG PET
 - Amyloid PET
 - Tau PET

NEUROTRANSMITTERS INVOLVED

- Acetylcholine: ↓ (Nucleus of Meynert)
- Noradrenaline: ↓ (Locus Coeruleus)
- Serotonin: ↓ (Dorsal raphe)

Which part of the brain is earliest to be involved in AD/ part most severely damaged?

1. Median temporal lobe (Entorhinal cortex) : earliest
2. Inferior temporal lobe
3. Lateral temporal lobe

CAUSES OF AD

CHROMOSOME 1	• Pre senilin 2 gene
CHROMOSOME 14	• Pre senilin 1 gene
CHROMOSOME 19	• Apo E gene • Allelic variants ○ Apo E₄ - ↑ risk ○ Apo E₂ - ↓ risk
CHROMOSOME 21	• APP (Amyloid precursor protein) extracopy gene → Aβ accumulation • Trisomy 21 : Pre-senile dementia

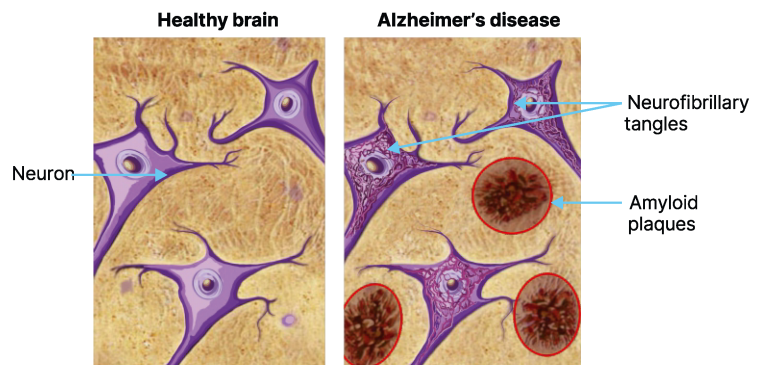
Important Information

Reduction of risk of AD

- NSAIDs
- Apo E₂

HISTOPATHOLOGY CORRELATION

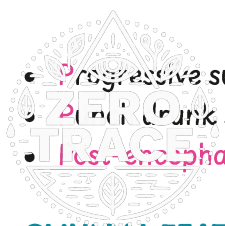
- Aβ- Amyloid/ Neuritic plaques
 - Extra neuronal finding
 - Aβ- Amyloid deposition : A/w dialysis dementia
- Neurofibrillary tangles
 - Intracytoplasmic finding
 - Composed of Tau_H protein → Damages neurons



CAUSES OF TAUPATHIES (Mnemonic : ACD- P4)

00:19:16

- Alzheimer's disease
- Corticobasal syndrome : Alien hand syndrome
- Down syndrome
- Pantothenate kinase associated neurodegeneration/ NBIA
 - MRI : eye of the tiger appearance



- Progressive supranuclear gaze palsy : retrocolis
- ~~Drunk~~ (Drunk syndrome (Dementia pugilistica)
- ~~L~~ ~~us~~ ~~e~~ ~~n~~ ~~c~~ ~~e~~ ~~p~~ ~~h~~ ~~a~~ ~~l~~ ~~i~~ ~~t~~ ~~i~~ ~~c~~ Parkinsonism

CLINICAL FEATURES

00:22:59

Mnemonic- 4 A

EARLY FEATURES

- ($\pm$) Seizures : GTCS/ focal seizures
- Mild cognitive impairment
 - Memory loss appreciated by patient
 - Can't track finances
 - Can't follow instructions
 - Can't drive (minor accidents)
 - Can't do shopping
- Amnesia (involvement of temporal lobe)
- Apraxia (involvement of non-dominant parietal lobe)
- Aphasia (involvement of dominant parietal lobe)
 1. Naming of common objects (initial loss)
 2. Comprehension
 3. Fluency loss
- Anosognosia : unawareness of deficit

WORK UP

00:32:32

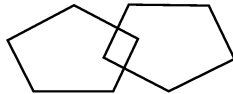
MINI - MENTAL STATE EXAMINATION (MMSE)

Patient's Name: _____ Date: _____

Instructions: Score one point for each correct response within each question or activity

MAXIMUM SCORE	PATIENT'S SCORE	QUESTIONS
5		"What is the year? Season? Date? Day? Month?"
5		"Where are we now? State? Country? Town/city? Hospital? Floor?"
3		The examiner names three unrelated objects clearly and slowly, then the instructor ask the patient to name al three of them. The patient's response is used for scoring. The examiner repeats them until patient learns all of them, if possible
5		"I would like you to count backward from 100 by sevens." (93, 86, 79, 72, 65, ...) Alternative: "Spell WORLD backwards." (D-L-R-O-W)
3		"Earlier I told you the names of three things. Can you tell me what those where?"
2		Show the patient two simple objects, such as a wristwatch and a pencil, and ask the patient to name them



1		"Repeat the phrase: 'No ifs, and, or buts'"
		"Take the paper in your right hand, fold it in half, and put it on the floor." (The examiner gives the patient a piece of blank paper)
1		"Please read this and do what it says." (Written instruction is "Close your eyes.")
1		"Make up and write a sentence about anything." (This sentence must contain a noun and a verb)
1		"Please copy this picture." (The examiner gives the patient a blank piece of paper and asks him/her to draw the symbol below. All 10 angles must be present and two must intersect) 
30		Total

• MMSE

- Evaluate dementia :30 items
- < 24 → dementia
- 21- 24 → mild
- 10 - 20 → moderate dementia
- < 10 → severe
- Depression can result in pseudodementia

• FDG Glucose PET

- Imaging modality
- ↓ metabolism of post parietal and temporal lobe

• Amyloid PET : Aβ neuritic plaques

• Tau PET : NFT

• CSF- Aβ₄₂ levels : ↓

- Biomarker of preclinical AD

• IOC : FDG PET / Amyloid PET

Important Information

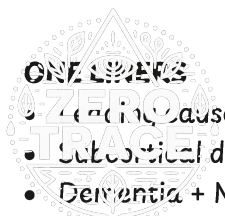
How to evaluate

- Working memory : (Spell a simple word in reverse)
- Episodic memory : 3 word recall
- Attention and calculation : serial 7 subtraction

TREATMENT

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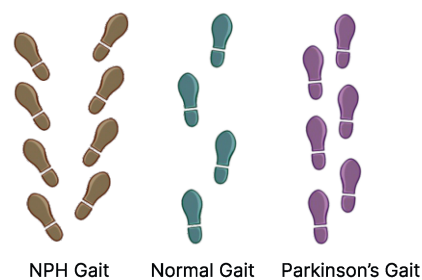
- Aducanumab
 - MOA: targets highly aggregated forms of Aβ amyloid
- Donepezil : DOC of AD
- Rivastigmine + Galantamine : Anticholinergic drug
- Tacrine : Hepatotoxic (withdrawn)



- Leading cause of Cortical dementia : AD
- Subcortical dementia : Parkinson disease
- Dementia + Myoclonus: VCJD
- Dementia + Chorea : Huntington chorea/ disease
- Presenile dementia : Down syndrome
- Reversible dementia:
 - Normal pressure hydrocephalus
 - Hypothyroidism
 - Vitamin B₁₂ deficiency
- Dementia with Lewy body
 - Visual hallucination
 - PD features
 - REM behavior disorder

NORMAL PRESSURE HYDROCEPHALUS

- Hakim triad (Wet, wobbly and wacky)
 - A.I.D
 - Ataxia (Magnetic gait)
 - Incontinence
- Fisher test
 - Reversible dementia
 - CSF removal (20-30 ml) → improves gait





Q. Clock face test is used for assessment of which of the following?

- a. Dementia
- b. Apraxia
- c. Neglect
- d. Socially inappropriate behaviour

Ans: (a)

Q. Which of the following is correct about Hakim's triad?

- a. Ataxia, HTN, Dementia
- b. HTN, Incontinence of urine and Dysphagia
- c. Ataxia, Incontinence of urine and Dementia
- d. Diplopia, Incoordination and Delirium

Ans: (c)



MULTIPLE SCLEROSIS

- Autoimmune demyelination predominantly involves the brain
- Main manifestations- **UMNL**
- CN II - m/c involved ↓ **Optic neuritis**
- **DIS** - Dissemination in space → Different topographic areas in the brain involved
- **DIT** - Dissemination in time → Development of lesions over a period of time

MRI	LOCATION	MRI	LOCATION
	• Periventricular plaques (Dawson fingers)		• Infratentorial plaque
	• Juxtacortical plaque		• Plaque in the Spinal cord

MacDONALD CRITERIA (NEW UPDATES)

00:03:48

- Imaging
 - **Central vein sign**
 - **Paramagnetic rim sign**
- CSF findings
 - **Oligoclonal bands** (locally synthesized antibodies against the myelin protein in the brain)
 - **CSF Kappa free light chains**

MULTIPLE SCLEROSIS

Autoimmune disease characterised by demyelination of the brain and spinal cord by intrathecally synthesized IgG that attacks Myelin basic protein (MBP) leading to following chain of events

1. Perivascular demyelination: Dawson finger

2. Axonal damage

○ Conduction block (↑ with ↑ in temp)

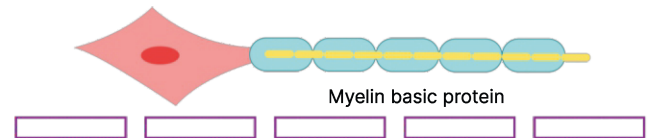
○ Temperature sensitive

○ **Uthoff phenomenon**: work out or hot shower in a patient can induce neurological manifestation

3. Oligodendrocytes

○ Remyelination axons → shadow plaques

4. Astrocytic gliosis



Important Information

- MBP reactive T cell attacks the endothelial cell in BV affects the integrity of BBB evidence of contrast uptake in the enhanced Gadonilum scan
- Natalizumab (monoclonal antibody) – prevents the recurrence of attacks from autoreactive MBP T-cells
- B-cells produce antibodies attacks MBP

COURSE OF DISEASE

• Relapsing MS (90% cases) :M/c type

• Primary progressive multiple sclerosis (10%)

• Secondary progressive multiple sclerosis

→ Combination of of RMS & PPMS

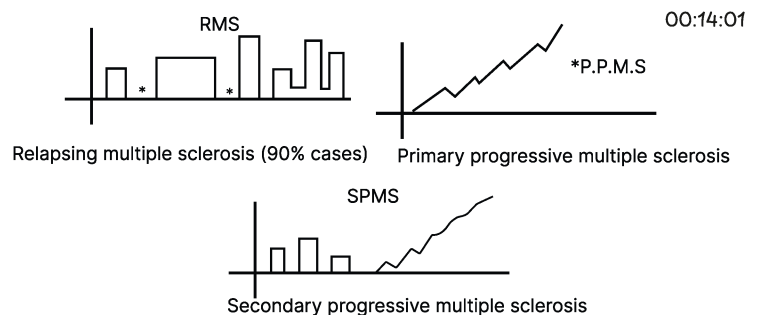
• Lesions occurring at different time: DIT

○ **Enhancing lesions** – acute or new lesions

○ **Non enhancing lesions** – older lesions

• Lesions occurring at different CNS locations: DIS

• More common in Caucasian / North american descent females and begins with optic neuritis



MULTIPLE SCLEROSIS	NMOSD	MOGAD
(-)	Anti AQP4 antibody	Anti- Myelin oligodendrocyte glycoprotein antibody disease
• White female • U/L ↓ in visual activity and color vision • Gradual onset • Internuclear ophthalmoplegia • One and half syndrome • Uthoff phenomenon	• Asian/ African descent • B/L optic neuritis (visual loss) • LETM: ≥ 3 contiguous spinal cord segments involved	• B/L optic neuritis • LETM (Conus medullaris) • Fundus: papillitis

CLINICAL FEATURES

• **Sensory symptoms** (37%)

○ M/c

○ Tingling sensation

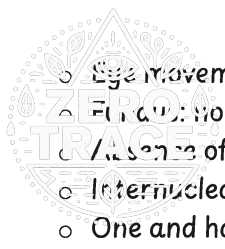
○ Pins and needle

• **Optic neuritis** (36%)

○ ↓ visual acuity/ dimness vision

○ ↓ color preception

00:23:45



- Eye movements causing pain
- Efferent: Normal/ papillitis/ optic atrophy
- Absence of direct light reflex/ Presence of consensual light reflex: Marcus Gunn pupil
- Internuclear ophthalmoplegia
- One and half syndrome
- Visual blurring d/t CN palsy
 - Ask the patient to cover each eye one by one
 - If visual blurring resolves: CN palsy
 - If visual blurring persists: optic neuritis
- Weakness of limbs
 - ↓ strength
 - ↓ dexterity
 - Exercise induced weakness
 - (+) Babinski sign (extensor plantars/ long tract signs)
- Facial nerve weakness
 - Resembles Bell's palsy
- Cerebellar involvement
 - Gait ataxia
 - Truncal ataxia
 - Dysaidokinesia
- Ancillary symptoms
 - Lhermitte's symptoms: neck flexion → pain in extremities
 - Trigeminal neuralgia
 - Glossopharyngeal neuralgia
 - Facial myokymia: twitching of eye muscles
 - Bladder involvement: Detrusor dyssynergia
 - Sexual complaints
 - Cognition issues, Epilepsy and falls
 - Uthoff phenomenon

DIRECT AND CONSENSUAL LIGHT REFLEX

00:33:35

- Normal pupillary reflex, when torched on eye
 - (+) DLR and (+) CLR
- In MS,
 - Light torched in affected eye: (-) DLR and (+) CLR – relative afferent pupillary defect
 - Light torched in the unaffected eye: (+) DLR and (+) CLR

FACIAL NERVE WEAKNESS

- No frowning (affected occipitofrontalis muscle)
- Drooping of face on the affected side
- Uprolling of the eyeball while closing the eye on the affected side: Bell's phenomenon

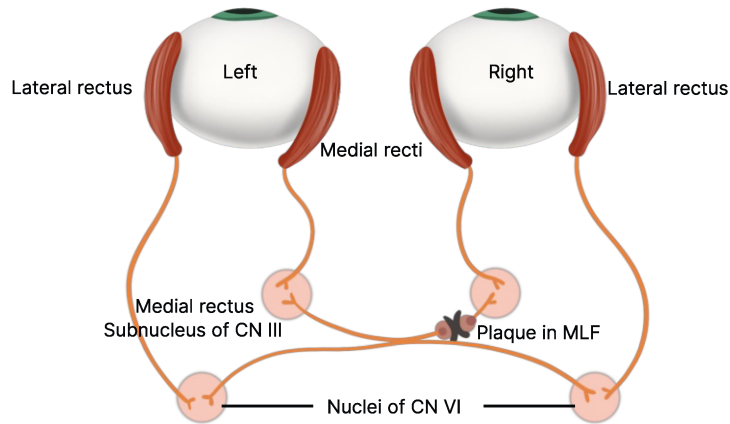


Left infranuclear 7th nerve palsy



INTERNUCLEAR OPHTHALMOPLÉGIA

• **Crossed** between 6th and 3rd CN of opposite sides occurs via Medial longitudinal fasciculus → horizontal gaze



- Plaque in MLF → medial recti muscle of one side not working → loss of adduction of the particular eye → ipsilateral impaired adduction

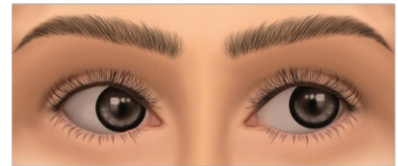


Normal



(+)

(+)



Nystagmus

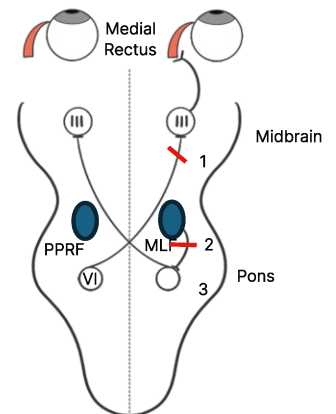
(-)

I/L Impaired adduction

- Rationale for nystagmus: to counterbalance the perceived overcorrection of contralateral eye/ abducting eye
- INO → involvement of MLF

ONE AND A HALF SYNDROME

- Plaque at the CN XI nucleus: **absence of abduction**
 - I/L Plaque at MLF : **absence of adduction**
 - Plaque at the CN XI nucleus (C/L eye) : **absence of adduction**
 - Only abduction on C/L eye is possible
- } Inability to move



EYE INVOLVEMENT IN MS

- Horizontal gaze palsy
- Internuclear ophthalmoplegia
 - MLF only involved
 - **I/L Impaired adduction**
 - **C/L nystagmus**
- One and half syndrome d/t involvement of
 - PPRF
 - MLF (Medial longitudinal fasciculus)
 - **Only C/L abduction possible**
- Eight and half syndrome (CN XII involvement)
- Pendular nystagmus
 - Cerebellar involvement

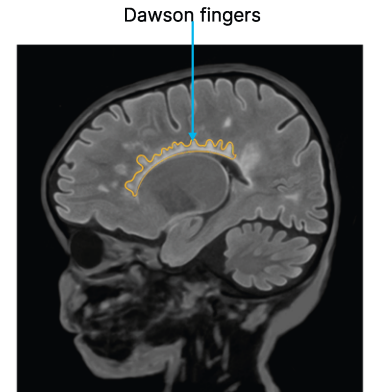
LINEARITY SIGN

- Sudden neck flexion can cause pain in the spine and extremities

REVISED McDONALD CRITERIA

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- > 2 attacks/ episodes of symptoms (plus)
 - > 2 objective clinical signs in non-contiguous areas of brain
 - If only 1 clinical sign is seen
 - 2nd involvement needs to be documented by MRI or visual evoked potential
 - Typical areas of involvement in MS
 - Juxtacortical area
 - Periventricular area
 - Infratentorial area
 - Spinal cord
- } ≥ 1 lesion in 2/4 areas
- MRI brain shows finger like projections in periventricular areas
 - Progressive course from onset > 6 months without superimposed relapses
 - Documentation of intrathecal IgG synthesis by CSF electrophoresis
 - Evidence oligoclonal bands (OCB)
 - IgG index: CSF IgG: Albumin / Serum IgG: Albumin

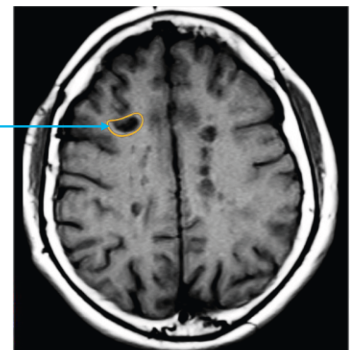


WORK-UP

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- Gadolinium-enhanced MRI
 - ↑ permeability BBB: demonstrated by Gadolinium enhancement
 - Plaques (residual)
 - Dawson fingers (T2)
 - Black hole (T1)
 - Central vein sign and paramagnetic rim sign
- Visual evoked potential: ↑ latency (Optic nerve involvement)
- Optical coherence tomography
- CSF IgG index, agarose electrophoresis: > 2 OCB and CSF Kappa free light chains
- Lumbar puncture: > 5 cells/microliter
 - If CSF protein > 1g, cells > 75/ ul – alternative diagnosis to be considered like neurosarcoidosis
- Other investigations for ruling out alternative diagnosis
 - ESR, Autoimmune panel
 - Serum Vitamin B₁₂ levels
 - Treponemal antibody

Black hole sign



EXPANDED DISABILITY SEVERITY SCORE: SEVERITY OF MS

- 3.5: fully ambulatory
- 9.0: helpless bedridden, can talk, can drink from straw and can eat
- 9.5: helpless bedridden, can't talk, can't drink, can't eat (CN IX, X, XII palsy)
- Benign MS: no neurological deficit but MRI lesions present

TREATMENT

- T/t of acute attack, initial demyelinating attack: IV methylprednisolone

DRUGS FREQUENTLY USED IN RMS

HIGHLY EFFECTIVE

- **Ocrelizumab**: deplete mature B cells
- **Ofatumumab**: S/C self injection monthly, new T2 lesions ↓ by 95%
- Rituximab
- **Natalizumab**: PML (Progressive Multifocal Leukoencephalopathy) risk caused by JC virus. So use only in JC virus **antibody-negative** patients

MODERATELY EFFECTIVE

- S₁P modulators
 - MOA: egress of lymphocytes from 2 lymphoid organs
 - Fingolimod, Ozanimod (oral drugs)
 - S/E: heart block and macular edema
- Dimethyl fumarate
 - MOA: modulation of release of proinflammatory cytokines

MODESTLY EFFECTIVE

- β-interferon
- Glatiramer: peptide copolymer of glutamic acid, lysine, alanine and tyrosine
 - Building blocks of Myelin
- Secondary Progressive MS: **Siponimod**
- Primary Progressive MS: **Ocrelizumab**

CLINICAL VARIANTS OF MS

01:01:45

- Marburg variant: Death 1-2 years

CLINICAL OUTCOMES:

- ↓ mean attack rate – **Natalizumab**
- ↓ change in disease severity – **Natalizumab**
- ↓ reduction in new T2 lesion on MRI – **Ofatumumab (S/C)**

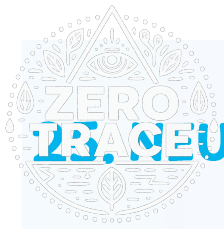
MCQs

01:03:04

Q. A 45 year old female presents to the neurology clinic with complaints of shooting pain radiating down her legs upon flexing her neck forward. On examination areflexia is noted lower extremities with Romberg sign and bilateral pupils that fail to constrict to light but accommodate. Diagnosis is?

- Tabes dorsalis
- Cervical spondylosis
- Multiple sclerosis
- SACD

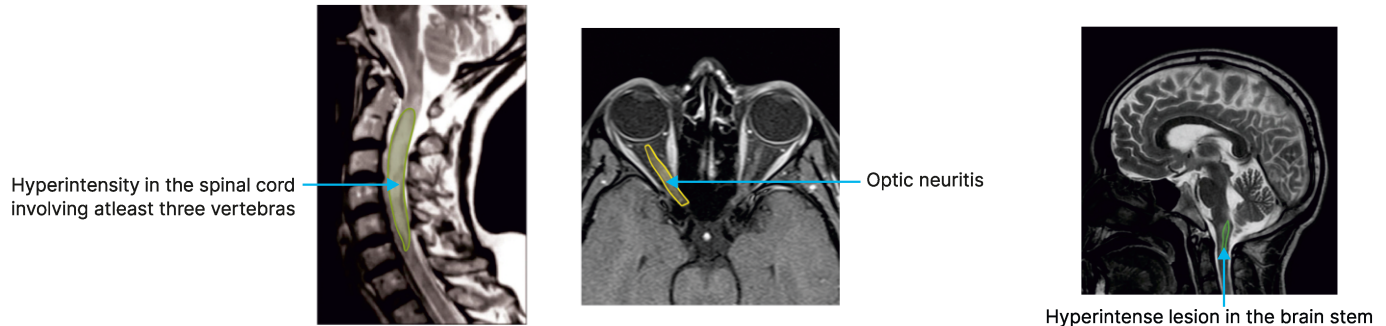
Ans. (a)



13. NEUROMYELITIS OPTICA SPECTRUM DISORDER

NEUROMYELITIS OPTICA SPECTRUM DISORDER

00:00:05

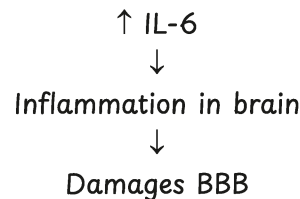


- Spinal cord involvement - longitudinally extensive transverse myelitis (>3 contiguous spinal cord segments involved)
- B/L optic neuritis → blindness
- MRI → cloud-like lesions in brainstem

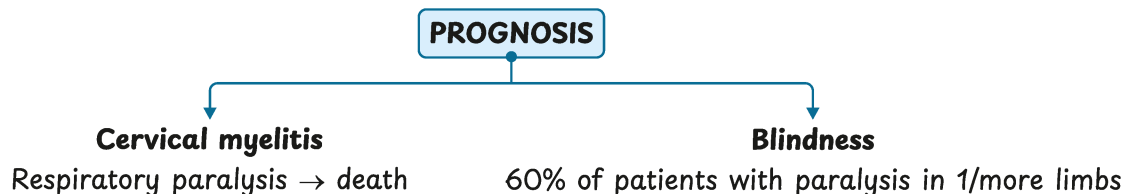
PATHOPHYSIOLOGY

00:03:05

- Astrocytopathy
 - Anti-aquaporin 4 antibody → activates complement system → complement mediated damage → damage to foot processes of astrocytes
- ↑ IL-6 in CSF



PROGNOSIS



CORE CLINICAL CHARACTERISTICS

00:05:24

- Area postrema syndrome (nausea, vomiting, hiccups)
- Brainstem syndrome manifestations
- Cerebral syndrome - Seizures, encephalopathy
- Diencephalic syndrome
 - Damage to hypothalamus
 - Significant weight loss